

Automated Waste Classification Using Convolutional Neural Networks

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produced annually— a figure projected to grow to 3.4 billion tonnes by 2050 [1]. Effective waste segregation at the source is the cornerstone of sustainable recycling, yet conventional manual sorting remains slow, inconsistent, and expensive. Human operators are susceptible to fatigue, and the sheer volume of modern waste streams makes manual intervention an increasingly untenable solution.

Deep learning, and Convolutional Neural Networks (CNNs) in particular, has transformed image classification across domains, offering an opportunity to automate the visual recognition of waste materials. Unlike hand crafted feature pipelines, CNNs learn hierarchical visual representations end-to-end, enabling robust generalisation to unseen waste images under varied lighting, orientations, and backgrounds.

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Abstract—Efficient waste management is a critical environmental challenge exacerbated by rapid urbanisation and population growth. Manual waste segregation is labour-intensive, error-prone, and unable to scale with the increasing volume of solid waste generated worldwide. This paper presents an automated waste classification system based on a custom Convolutional Neural Network (CNN) trained on the publicly available *Garbage Classification v2* dataset from Kaggle, containing 12,259 labelled images across ten categories: battery, biological, cardboard, clothes, glass, metal, paper, plastic, shoes, and trash. Images are resized to 128×128 pixels and pixel values normalised to $[0, 1]$ prior to training. The CNN architecture consists of three successive convolutional blocks followed by fully connected layers with dropout regularisation, using Sparse Categorical Cross-Entropy loss and the Adam optimiser over ten epochs. The model achieves a final validation accuracy of 68.09%, with per-class F1-scores ranging from 0.43 (metal) to 0.86 (clothes). The results confirm both the feasibility of CNN-based automated waste sorting and the need for data augmentation and transfer learning to improve generalisation, providing a foundation for practical AI-driven waste management solutions.

Index Terms—Convolutional Neural Network, Waste Classification, Deep Learning, Image Classification, Computer Vision, TensorFlow

I. INTRODUCTION

A. Background and Motivation

Global solid waste generation has reached alarming levels, with the World Bank estimating that over two billion tonnes of municipal solid waste are

B. Problem Statement

Manual waste segregation leads to high rates of recyclable material being incorrectly discarded to landfill, increasing greenhouse gas emissions and reducing recycling efficiency. The absence of scalable, low-cost automated sorting mechanisms is a significant barrier to circular economy practices.

C. Objectives

The primary objectives of this project are:

- Develop a CNN-based model that classifies waste images into ten distinct categories.
- Automate the waste segregation process to reduce dependence on manual sorting.
- Evaluate model performance using standard classification metrics and per-class analysis.
- Identify limitations and propose concrete directions for improvement.

D. Why a CNN-Based Approach

CNNs automatically learn discriminative spatial features—edges, textures, shapes—from raw pixel data without requiring manually engineered descriptors. This makes them particularly effective for visual material recognition, where distinguishing plastic from glass or metal relies on subtle surface and colour cues that are difficult to capture with rule-based methods [7].

II. DATASET AND PREPROCESSING

A. Dataset Description

The project uses the *Garbage Classification v2*

dataset sourced from Kaggle [2]. The dataset is organised into class-wise subdirectories within a standardized_384 subfolder, which was automatically detected at runtime. A summary of the dataset properties is provided in Table I.

The ten waste categories, along with the number of samples in the validation split, are listed in Table II.

B. Data Loading

The dataset is loaded using TensorFlow's `image_dataset_from_directory` API with a fixed random seed of 123, a validation split of 20%, and a batch size of 32. Labels are automatically inferred from directory names as integer indices.

TABLE I: Dataset Summary

Property	Value
Source	Kaggle (Garbage Classification v2) Subfolder
Used standardized_384	
Total Images	12,259
Number of Classes	10
Training Split	9,808 images (80%)
Validation Split	2,451 images (20%)

TABLE II: Class Distribution in Validation Set

Class	Validation Samples
Battery	149
Biological	123
Cardboard	286
Clothes	396
Glass	324
Metal	189
Paper	270
Plastic	330
Shoes	301
Trash	83
Total	2,451

C. Preprocessing Pipeline

The following preprocessing steps are applied before training:

- 1) Image Resizing: All images are resized to a uniform resolution of 128×128 pixels.
- 2) Normalisation: Pixel intensity values in the range [0, 255] are rescaled to [0, 1] by applying a `Rescaling(1./255)` layer to both training and validation sets.
- 3) Dataset Splitting: An 80%/20% training-validation split is used, yielding 9,808 training images and 2,451 validation images.

No online data augmentation (flipping, rotation, zoom) was applied during this experimental run, which, as discussed in Section V, contributes to the observed overfitting in later epochs.

III. SYSTEM DESIGN AND METHODOLOGY A.

Model Architecture

1) *Custom CNN Architecture*: The baseline model is a custom CNN designed for multi-class waste image classification. The architectural components are as follows:

- Convolutional Blocks: Three successive convolu-

tional blocks are used. The first block contains a Conv2D layer with 32 filters, the second block contains a Conv2D layer with 64 filters, and the third block contains a Conv2D layer with 128 filters. Each convolutional layer uses a 3 × 3 kernel and ReLU activation, followed by a 2 × 2 Max-Pooling layer.

- Flatten Layer: Converts the extracted feature maps into a one-dimensional feature vector.
- Fully Connected Layer: A dense layer with 128 units and ReLU activation is used for classification learning.
- Dropout Layer: A Dropout layer with a rate of 0.5 is used to reduce overfitting.
- Output Layer: A dense layer with 10 units and Softmax activation is used for predicting one of the ten waste categories.

The full architecture is illustrated in Fig. 1.

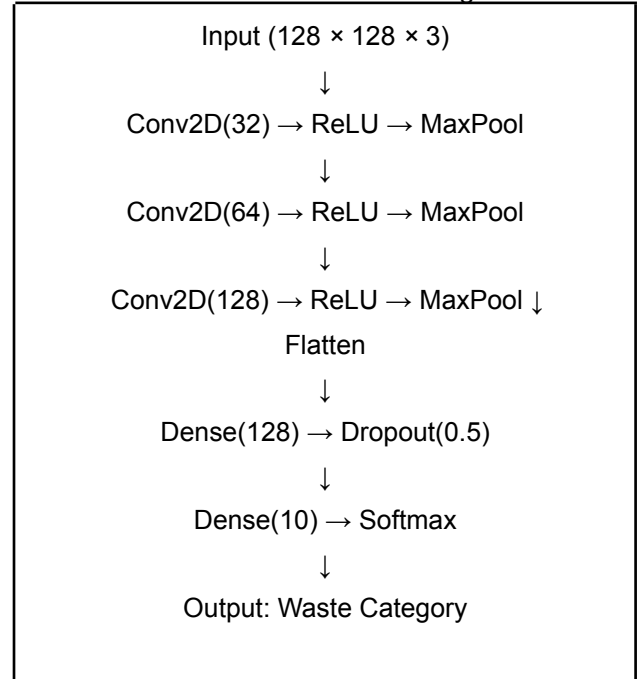


Fig. 1: Custom CNN architecture for 10-class waste classification.

B. Training Procedure and Hyperparameters

The training configuration is summarised in Table III.

TABLE III: Training Hyperparameters

Hyperparameter	Value
Loss Function	Sparse Categorical Cross-Entropy
Optimiser	Adam (default $\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning Rate	1×10^{-3} (default Adam)
Batch Size	32
Epochs	10
Input Resolution	128 × 128
Normalisation	Rescaling by 1/255
Dropout Rate	0.5
Random Seed	123

The model is compiled with

sparse_categorical_crossentropy as the loss function (compatible with integer class labels), the Adam optimiser, and accuracy as the evaluation metric. Training is run for a fixed 10 epochs without early stopping or learning rate scheduling.

C. Tools and Frameworks

- Programming Language: Python 3.12
- Deep Learning Framework: TensorFlow / Keras
- Data Ingestion: kagglehub for dataset download
- Evaluation: scikit-learn (classification_report, confusion_matrix)
- Visualisation: Matplotlib, Seaborn
- Development Environment: Google Colaboratory (GPU: NVIDIA T4, 16 GB VRAM)

IV. IMPLEMENTATION AND RESULTS

A. Experimental Setup

All experiments were conducted on Google Colaboratory using a single NVIDIA T4 GPU. The dataset was downloaded via kagglehub and the standardized_384 subfolder was automatically located at runtime. Labels were inferred directly from the class-wise folder structure. The complete Colab notebook is available at: https://colab.research.google.com/drive/1p9EKSOIMaph0k-u3B_zs_Bko2hPNiAZd.

B. Training Performance

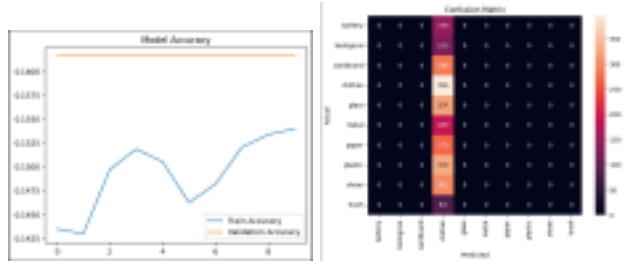
Table IV presents the per-epoch training and validation metrics recorded over the 10 training epochs.

TABLE IV: Per-Epoch Training History

Epoch	Tr. Loss	Tr. Acc.	Val. Loss	Val. Acc.
1	1.8861	32.87%	1.5461	48.88%
2	1.4953	48.17%	1.3230	55.53%
3	1.3219	55.03%	1.2298	58.67%
4	1.1669	60.01%	1.1446	61.89%
5	1.0709	63.47%	1.1479	62.63%
6	0.9407	67.94%	1.1243	63.77%
7	0.8571	70.20%	1.1384	64.87%
8	0.7591	73.98%	1.1720	67.16%
9	0.6661	76.68%	1.1518	66.99%
10	0.6165	78.32%	1.2028	68.09%

The training accuracy improves steadily from 32.87% (Epoch 1) to 78.32% (Epoch 10). The validation accuracy shows consistent improvement through the early epochs, reaching 68.09% by Epoch 10. However, from Epoch 5 onwards, the validation loss begins to plateau and slightly increase while the training loss continues to decline, indicating the onset of overfitting.

C. Training Result



(a) Model Accuracy (b) Confusion Matrix

Fig. 2: Training performance metrics

D. Final Evaluation Metrics

The trained model is evaluated on the held-out validation set. The final validation accuracy is 68.09%. Per-class precision, recall, and F1-score are reported in Table V.

E. Class-Level Analysis

The model achieves its best performance on the *clothes* class (F1 = 0.86, recall = 0.90), likely due to its distinctive textures and patterns compared to other categories. *Battery* and *cardboard* also achieve strong F1-scores of 0.73, benefiting from their consistent visual appearance across samples.

The lowest F1-score is observed for *metal* (0.43) and *trash* (0.44). The metal class has the second-lowest recall (0.36), indicating frequent confusion with visually similar categories such as glass or plastic. The *trash* class, comprising only 83 validation samples, suffers from both

TABLE V: Per-Class Classification Report on Validation Set

Class	Prec.	Rec.	F1	Support
Battery	0.77	0.69	0.73	149
Biological	0.64	0.73	0.68	123
Cardboard	0.74	0.72	0.73	286
Clothes	0.83	0.90	0.86	396
Glass	0.60	0.73	0.66	324
Metal	0.54	0.36	0.43	189
Paper	0.60	0.57	0.59	270
Plastic	0.63	0.62	0.63	330
Shoes	0.67	0.74	0.71	301
Trash	0.72	0.31	0.44	83
Macro Avg	0.68	0.64	0.65	2,451
Weighted Avg	0.68	0.68	0.67	2,451
Overall Accuracy 68.09%				

class imbalance and high intra-class visual heterogeneity, as “trash” encompasses a diverse range of mixed-waste objects.

V. DISCUSSION

A. Interpretation of Results

The custom CNN achieves a validation accuracy of 68.09% on a 10-class waste classification task, which is a meaningful result for a compact architecture trained without data augmentation or pre-trained weights. The steady divergence between training

accuracy (78.32%) and validation accuracy (68.09%) at Epoch 10 confirms the presence of overfitting, which begins to manifest from approximately Epoch 5, as evidenced by the rising validation loss despite a continuously decreasing training loss.

The per-class results reveal that the model captures high-level visual features effectively for visually distinctive categories (clothes, battery, cardboard), but struggles with materials that share surface appearance—such as metal and glass—or categories that are highly heterogeneous, such as trash.

B. Limitations and Challenges

Absence of Data Augmentation: No augmentation techniques (flipping, rotation, zoom, brightness adjustment) were applied during training. Augmentation is a key regularisation strategy that artificially expands the training set and reduces overfitting, and its absence is likely the primary driver of the training–validation accuracy gap observed from Epoch 5 onwards.

Low Input Resolution: Images were resized to 128×128 pixels. A higher resolution (e.g., 224 × 224) would pre-serve finer surface details and material textures, potentially improving classification accuracy, especially for visually similar categories like metal and plastic.

Class Imbalance: The validation set contains only 83 trash samples compared to 396 clothes samples—a ratio of nearly 5:1. No class-weighting or oversampling strategy was employed, which disadvantages underrepresented categories during training.

No Transfer Learning: The model was trained from scratch without leveraging features from large-scale pre-trained models such as MobileNetV2 [3] or ResNet50 [4]. Transfer learning has been shown to substantially improve performance on moderately sized domain-specific datasets.

Fixed Training Duration: Training was conducted for a fixed 10 epochs without early stopping or learning rate scheduling. This prevents the model from automatically halting before overfitting becomes pronounced.

Mixed Waste: The current approach assumes one dominant waste type per image. Real-world waste often consists of composite materials (e.g., a plastic bottle with a paper label), which the single-label model cannot handle.

C. Possible Improvements

- Apply data augmentation (horizontal/vertical flips, random rotation of $\pm 30^\circ$, zoom, brightness and contrast jitter) exclusively on the training set to reduce overfitting.
- Increase input resolution to 224×224 and benchmark performance.
- Integrate transfer learning with MobileNetV2 [3] or ResNet50 [4] for improved accuracy with limited domain-specific data.

- Add Batch Normalisation after each convolutional layer to stabilise training and further regularise the network.
- Implement early stopping with patience and a ReduceLROnPlateau scheduler to prevent overfitting and accelerate convergence.
- Apply class-weighted loss or oversampling (e.g., SMOTE on embeddings) to address the class imbalance affecting trash, biological, and battery categories.
- Explore lightweight architectures such as EfficientNet [5] or MobileNetV3 for real-time deployment on edge hardware.
- Investigate object detection frameworks (e.g., YOLOv8 [6]) to handle images containing multiple waste items simultaneously.

VI. CONCLUSION

This paper presented an automated waste classification system based on a custom Convolutional Neural Network trained on the Garbage Classification v2 dataset from Kaggle. Using 12,259 labelled images across ten categories—battery, biological, cardboard, clothes, glass, metal, paper, plastic, shoes, and trash—the model was trained for 10 epochs using the Adam optimiser and Sparse Categorical Cross-Entropy loss, with input images resized to 128×128 pixels and normalised to [0, 1].

The key findings are:

- The model achieves a final validation accuracy of 68.09%, demonstrating the viability of CNN-based approaches for multi-class waste classification.
- Per-class F1-scores range from 0.43 (metal) to 0.86 (clothes), with underrepresented and visually ambiguous classes showing weaker performance.
- The training–validation accuracy divergence (78.32% vs. 68.09% at Epoch 10) indicates overfitting, attributable to the absence of data augmentation and small training set size for certain classes.
- The *clothes* category was classified most reliably, while *metal* and *trash* presented the greatest classification challenge.

Future work will focus on: (i) integrating data augmentation and transfer learning (MobileNetV2, ResNet50) to

close the performance gap; (ii) increasing input resolution and applying early stopping; (iii) addressing class imbalance through weighted loss functions; and (iv) deploying an optimised model on edge hardware (e.g., Raspberry Pi, NVIDIA Jetson) for real-time waste sorting. The proposed system has clear potential to contribute meaningfully to sustainable waste management and circular economy initiatives.

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